Lab-4 (String)

1. Write a c-program to input a string and print it.

Ans:

```
#include<stdio.h>
#include<string.h>
int main()
{
	char a[100];
	printf("Enter the string:");
	fgets(a,100,stdin);
	a[strlen(a)-1]='\0';
	printf("The inputed string is:%s\n",a);
	return 0;
}
```

Output: C:\Users\aayus\OneDrive\Desktop\New folder\C-program\string.exe

```
Enter the string:Aayush
The inputed string is:Aayush

--------------------------------
Process exited after 13.32 seconds with return value 0
Press any key to continue . . .
```

2. Write a c-program to find length of a string without using string handling function.

Ans:

```
#include<stdio.h>
#include<string.h>
int main()
{
	char a[50];
	printf("Enter the string:");
	fgets(a,50,stdin);
	int i,length;
	while(a[i]!='\0')
	{
		if(a[i]=='\n')
		{
			a[i]='\0';
			break;
		}
```

```
            i++; }
        length=strlen(a);
        printf("The inputed string is:%d\n",length);
        return 0;
}
```

Output: C:\Users\aayus\OneDrive\Desktop\New folder\C-program\string.exe

```
Enter the string:Ram
The inputed string is:3

--------------------------------
Process exited after 2.824 seconds with return value 0
Press any key to continue . . .
```

3. Write a c-program to find length of a string using string handling function.

Ans:

```
#include<stdio.h>
#include<string.h>
int main()
{
        char a[50];
        printf("Enter the string:");
        fgets(a,50,stdin);
        int i,length;
        a[strlen(a)-1]='\0';
        length=strlen(a);
        printf("The inputed string is:%d\n",length);
        return 0;
}
```

Output: C:\Users\aayus\OneDrive\Desktop\New folder\C-program\string.exe

```
Enter the string:Shyam
The inputed string is:5

--------------------------------
Process exited after 3.233 seconds with return value 0
Press any key to continue . . .
```

4. Write a c-program to finding no. of alphabet in a string without using string handling function.

Ans:

```
#include<stdio.h>
#include<string.h>
int main()
{
	char a[50];
	int i=0,count=0;
	printf("Enter the string:");
	fgets(a,50,stdin);
	while(a[i]!='\0')
	{
		if(a[i]=='\n')
		{
			a[i]='\0';
			break;
		}
		i++;
	}
	i=0;
	while(a[i]!='\0')
	{
		if(a[i]!=' ' && a[i]!='\t')
		{
			count=count+1;
		}
		i++;
	}
	printf("The length of inputed string is:%d",count);
	return 0;
}
```

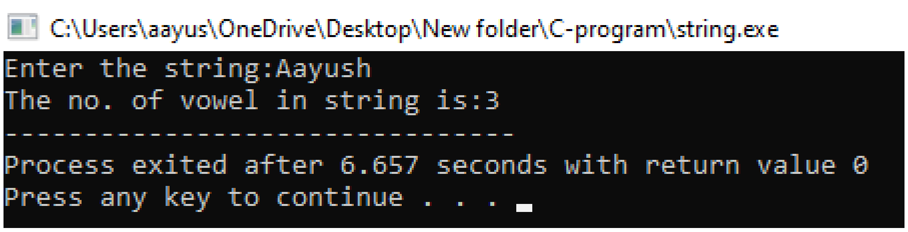
Output: C:\Users\aayus\OneDrive\Desktop\New folder\C-program\string.exe

```
Enter the string:Ram Ram
The length of inputed string is:6
--------------------------------
Process exited after 5.61 seconds with return value 0
Press any key to continue . . .
```

5. Write a c-program to count vowel in string.

Ans:

```
#include<stdio.h>
#include<string.h>
int main()
{
	char a[50];
	printf("Enter the string:");
	fgets(a,50,stdin);
	a[strlen(a)-1]='\0';
	int i=0,count=0;
	strupr(a);
	while(a[i]!='\0')
	{
		if(a[i]=='A' || a[i]=='E' || a[i]=='I' || a[i]=='O' || a[i]=='U' )
		{
			count=count+1;
		}
		i++;
	}
	printf("The no. of vowel in string is:%d",count);
	return 0;
}
```

Output: 

6. Write a c-program to copy input string to another.

Ans:

```
#include<stdio.h>
#include<string.h>
int main()
{
	char a[50],b[50];
	printf("Enter the string:");
	fgets(a,50,stdin);
	a[strlen(a)-1]='\0';
	strcpy(b,a);
	printf("The copy of inputed string is:%s",b);
	return 0; }
```

Output: C:\Users\aayus\OneDrive\Desktop\New folder\C-program\string.exe

```
Enter the string:ram
The copy of inputed string is:ram
--------------------------------
Process exited after 1.613 seconds with return value 0
Press any key to continue . . .
```

7. Write a c-program to concatenate (ie. add) two string.

Ans:

```
#include<stdio.h>
#include<string.h>
int main()
{
        char a[50],b[50];
        printf("Enter the string:");
        fgets(a,50,stdin);
        printf("Enter the string:");
        fgets(b,50,stdin);
        a[strlen(a)-1]='\0';
        b[strlen(b)-1]='\0';
        strcat(a,b);
        printf("The copy of inputed string is:%s",a);
        return 0;
}
```

Output: C:\Users\aayus\OneDrive\Desktop\New folder\C-program\string.exe

```
Enter the string:Aayush
Enter the string:Baral
The copy of inputed string is:AayushBaral
--------------------------------
Process exited after 9.84 seconds with return value 0
Press any key to continue . . .
```

8. Write a c-program to check whether two string are equal or not.

Ans:

```
#include <stdio.h>
#include <string.h>
int main()
{
   char str1[50], str2[50];
   printf("Enter the first string: ");
   fgets(str1,50,stdin);;
   printf("Enter the second string: ");
   fgets(str2,50,stdin);
```

```
    if (strcmp(str1, str2) == 0)
    printf("\nThe strings are equal.\n");
    else
    printf("\nThe strings are not equal.\n");
    return 0;
}
```

Output: C:\Users\aayus\OneDrive\Desktop\New folder\C-program\string.exe

```
Enter the first string:Aayush
Enter the second string:Aayush
The strings are equal
--------------------------------
Process exited after 5.76 seconds with return value 0
Press any key to continue . . .
```

9. Write a c-program to convert lowercase to upercase and viceversa.

Ans:

```
#include<stdio.h>
#include<string.h>
int main()
{
        char a[50];
        char b[50];
        printf("Enter string in lowercase:");
        scanf("%50[^\n]s",a);
        printf("Enter string in uppercase:");
        scanf(" %50[^\n]s",b);

        strupr(a);
        strlwr(b);
        printf("The lower case entered text into uppercase is: %s\n",a);
        printf("The upper case entered text into lowercase is: %s",b);
        return 0;
}
```

Output: C:\Users\aayus\OneDrive\Desktop\New folder\C-program\string.exe

```
Enter string in lowercase:ram
Enter string in uppercase:SITA
The lower case entered text into uppercase is: RAM
The upper case entered text into lowercase is: sita
--------------------------------
Process exited after 14.62 seconds with return value 0
Press any key to continue . . .
```

10. Write a c-program to check whether the input string is palindrome or not.

Ans:

```
#include<stdio.h>
#include<string.h>
int main()
{
	char a[50];
	char b[50];
	printf("Enter any sting for check:");
	scanf("%50[^\n]s",a);
	strcpy(b,a);
	strrev(a);
	if(strcmp(a,b)==0)
	printf("It is palindrone string");
	else
	printf("It is not palindrone string");
	return 0;
}
```

Output: C:\Users\aayus\OneDrive\Desktop\New folder\C-program\string.exe

```
Enter any sting for check:madam
It is palindrone string
--------------------------------
Process exited after 48.26 seconds with return value 0
Press any key to continue . . .
```